Lab 3 – Virtual Camera

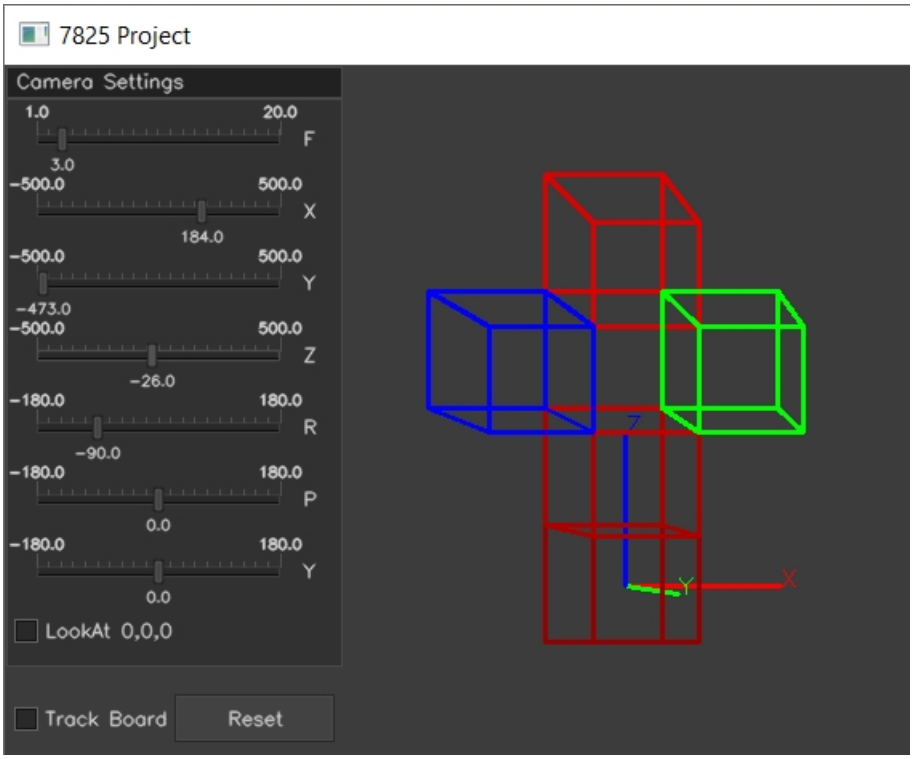
1 Introduction

In this lab a webcam will draw 3D boxes in the ELEX78255 Template project and visualize the environment by rendering through a virtual camera using the perspective (pin-hole) camera model.

2 Experiment

A) Download the ELEX7825 Template from GITHUB and the opencv.zip file from D2L. Unzip the OpenCV library to the template folder and ensure that it will compile.

B) Draw 6 cubic boxes with 5 cm sides to form a virtual robot as shown below. Also draw the world coordinate system origin at the center of the lowest cube. Positive X is towards the right, positive Z is towards the top and positive Y is into the screen.



C) Add a virtual camera object which implements a perspective pin-hole camera. Determine the intrinsic and extrinsic camera matrixes using the following default values: image size of 1280x720, or the size of your webcam image, a critical point in the middle of the image, a focal length of 3 mm. Position the camera at X = 0 mm, Y = -500 mm, Z = 0 mm. Note the camera is pointing along the +Z direction and so you must orient the pitch (rotate around X) by -90 to bring the robot into view.

D) Link the trackbars X / Y / Z / Roll / Pitch / Yaw / Focus parameters to the virtual camera so changes in the trackbar values are automatically reflected in the virtual image.

3 OPTIONAL

Implement the LookAt (0,0,0) function where you can move the X, Y, Z position of the camera and the R/P/Y automatically update to point towards the origin (0, 0, 0).